

Ibn Tofail University

Analysis V

Problem Set IV

Exercise 1:

For $(x, y) \in \mathbb{R}^2$, we define

$$f(x, y) = \begin{cases} xy \sin\left(\frac{\pi}{2} \cdot \frac{x^2 - y^2}{x^2 + y^2}\right) & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

1. Study the continuity of f .
2. Study the differentiability and determine if f is of class C^1 .
3. Study the second-order partial derivatives and determine if f is of class C^2 .

Correction

Exercise 2:

Let f be a function from $\mathbb{R}^2$ to $\mathbb{R}$, defined by:

$$f(x, y) = \frac{x^2 y^2}{x^2 + y^2} \text{ if } (x, y) \neq (0, 0) \text{ and } f(0, 0) = 0$$

1. Study the continuity of f at $(0, 0)$.
2. Calculate the partial derivatives of f at $(0, 0)$.
3. Study the differentiability of f at $(0, 0)$.
4. Let $(x, y) \neq (0, 0)$. Calculate the partial derivatives of f at (x, y) .
5. Is f of class C^1 ?
6. Deduce the answer to (3) from (5).

Correction

Exercise 3:

Let f be a C^1 mapping from $\mathbb{R}^n$ to $\mathbb{R}$. We say that f is positively homogeneous of degree r (where r is a given real number) if and only if $\forall \lambda \in]0, +\infty[, \forall x \in \mathbb{R}^n, f(\lambda x) = \lambda^r f(x)$.

Show that for such a function, the following Euler identity holds:

$$\forall x = (x_1, \dots, x_n) \in \mathbb{R}^n, \quad \sum_{i=1}^n x_i \frac{\partial f}{\partial x_i}(x) = r f(x).$$

Correction

Exercise 4:

Let f be a function from $\mathbb{R}^2$ to $\mathbb{R}$. Let $(a, b) \in \mathbb{R}^2$. Suppose that f is differentiable in a neighborhood of (a, b) . Let S be the surface with equation $z = f(x, y)$.

1. Give the equation of the tangent plane P to S at the point $(a, b, f(a, b))$.
2. Give two direction vectors $\mathbf{u}$ and $\mathbf{v}$ of P and the normal vector to P .

Application:

Let $f(x, y) = \exp(x \ln(x^2 + y^2))$ for $(x, y) \neq (0, 0)$. Give the equation of the tangent plane to the surface $S = \{(x, y, z) \in \mathbb{R}^3 \mid z = f(x, y)\}$ at the point $(1, 0, f(1, 0))$.

Additional question:

Determine the Taylor expansion of f to order 3 at $(0, 1)$ using the uniqueness of a Taylor expansion and certain restrictions of f .

Correction

Exercise 5:

Find the extrema of the following functions:

1. $f(x, y) = x^3 + 3x^2y - 15x - 12y$
2. $f(x, y) = -2(x - y)^2 + x^4 + y^4$

Correction

Exercise 6:

Find the minimum of $f(x, y) = \sqrt{x^2 + (y - a)^2} + \sqrt{(x - a)^2 + y^2}$, where a is a given real number.

Correction**Exercise 7:**

Solve the following partial differential equations:

1. $2\frac{\partial f}{\partial x} - \frac{\partial f}{\partial y} = 0$ by setting $u = x + y$ and $v = x + 2y$.
2. $x\frac{\partial f}{\partial y} - y\frac{\partial f}{\partial x} = 0$ on $\mathbb{R}^2 \setminus \{(0, 0)\}$ by using polar coordinates.
3. $x^2\frac{\partial^2 f}{\partial x^2} + 2xy\frac{\partial^2 f}{\partial x\partial y} + y^2\frac{\partial^2 f}{\partial y^2} = 0$ on $]0, +\infty[\times \mathbb{R}$ by setting $x = u$ and $y = uv$.

Correction**Exercise 8:**

Let f be a real-valued function of a real variable defined in a neighborhood of a point a such that $f''(a)$ exists.

1. Let $\varepsilon > 0$ be a given real number. Show that for t sufficiently close to a , we have the inequality:

$$|f'(t) - f'(a) - (t - a)f''(a)| < \varepsilon|t - a|$$

2. Define $H(x) = f(x) - (x - a)f'(a) - \frac{(x - a)^2}{2}f''(a)$.

Deduce a bound for $H(y) - H(x)$ when x and y are sufficiently close to a .

3. Define:

$$F(x, y) = \frac{f(y) - f(x)}{y - x} \text{ for } y \neq x \text{ and } F(x, x) = f'(x)$$

Show that F is differentiable at the point (a, a) and that:

$$dF_{(a,a)} = \frac{1}{2}f''(a)(dx + dy)$$

Correction